

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 3: Classification

Ch 3 – Classification – Numerical Problems & Solutions

167 numerical marks, 8.6% of the archive, 40% of this chapter · 6 distinct methods · 12 PYQ instances solved · every answer independently recomputed. Decision-tree and Naive-Bayes figures cross-checked against Han & Kamber 3rd ed Examples 8.1–8.4, which use the same dataset the IOE papers copy.

Methods

- **M1** Decision tree by information gain (ID3) 79 Bh, 78 Bh, 76 Ch
- **M2** Gini index 75 Ch, 71 Ch
- **M3** Naive Bayes classification 72 Ch, 79 Bh
- **M4** k -Nearest Neighbour 80 Bh
- **M5** Confusion matrix metrics **8 papers**
- **M6** ANN gradients (backpropagation) 76 Ch

Every problem below ends with its solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**. The table says what each move is and where it is used.

The moves every chapter-3 numerical below is built from

Move	What it says (where: problem, step; orange = key step)	Move	What it says (where: problem, step; orange = key step)
entropy	Info = $-\sum p \log_2 p$ of the class counts at a node. 1.4 1	k nearest	Sort the distances and keep the k smallest. 4.1 2
gain	Info before minus weighted Info after the split; the largest wins. 1.4 2	majority vote	The class most common among the k nearest is the prediction. 4.1 3
pure branch	A partition holding one class has Info (or Gini) 0 and becomes a leaf. 1.4 3; Gini 3	read the axes	Find which axis is Actual before naming TP, FN, FP, TN. Papers flip it. 5.1 1; 5.2 1; 5.3/5.4 1; 5.5 1; 5.6 1; 5.7 1; 5.9 1; 5.10 1; 5.8 2
Gini	$1 - \sum p^2$: the impurity Gini-based trees minimise. Gini 1	tally	Compare actual and predicted position by position and count each outcome. 5.8 1
binary split	Gini splits in two: group the values into two subsets, try each grouping. Gini 2, 3	diagonal	Accuracy uses the diagonal (TP + TN) over everything. 5.1 2; 5.2 2; 5.3/5.4 2; 5.5 2; 5.9 2; 5.10 4; 5.8 3
prior	$P(C)$ = class count / all tuples. 3.1 1; 3.2 1	off-diagonal	Error rate uses the off-diagonal (FP + FN) over everything. 5.6 2; 5.9 3
class count	$P(x_k C)$ = tuples of class C with that value / tuples of class C . 3.1 2; 3.2 2	actual total	TPR, specificity and FPR divide by an ACTUAL class total, P or N . 5.1 3,5; 5.2 3,4; 5.3/5.4 3,4; 5.5 3,4; 5.6 3,4,5; 5.7 2,3; 5.9 4,5; 5.10 2,3; 5.8 5,6
Laplace	A zero count wipes out the product: add 1 to each count, k to the total. 3.2 3	predicted total	Precision divides by the PREDICTED-positive total, TP + FP. 5.1 4; 5.2 5; 5.3/5.4 5; 5.5 5; 5.7 4; 5.9 6; 5.8 4
Gaussian	A continuous attribute: fit μ , σ per class, read the density at x . 3.2 4	weighted sum	A neuron adds its weighted inputs, $net = \sum w_i x_i$. 6.1 1
product	Naive independence: multiply the per-attribute probabilities. 3.1 3; 3.2 5	sigmoid	$\sigma(z) = 1/(1 + e^{-z})$, and $\sigma' = \sigma(1 - \sigma)$. 6.1 1, 3
largest wins	Compare the candidates; the largest score is the answer. Gini 4; 3.1 4; 3.2 5	chain rule	Multiply the local derivatives along the one path from the loss to the weight. 6.1 2, 3
distance	Euclidean distance from the query to every training tuple. 4.1 1		

M1. Decision tree by information gain (ID3)

Two things to know before you start.

- **Problems 1.2 and 1.4 are different tables.** Same ten-row transport layout, same class counts, but four rows differ. The root is Travel cost in both, yet the runner-up is not: Income level (0.6955) in 1.2, a dead tie between Car ownership and Income level (0.5345) in 1.4. Never copy one answer into the other.
- **Two of the four tables have three classes** (1.2 and 1.4: Bus, Car, Train; 1.3: high, moderate, low). *Info* then has three terms and can exceed 1; the maximum for three classes is $\log_2 3 = 1.585$.

The recipe

- **Step 1.** Count the classes in the whole set D , then $Info(D) = -\sum_i p_i \log_2 p_i$, one term per class.
- **Step 2.** For each attribute A , split D by its values and fill *one row per value* of the table below: the class counts, that

branch's $Info(D_j)$, and the branch weighted by its share of D .

- **Step 3.** Sum the weighted column. That is $Info_A(D)$, the **residual** entropy after splitting on A .
- **Step 4.** $Gain(A) = Info(D) - Info_A(D)$. Compare the gains; the largest wins and becomes the node's test.
- **Step 5.** Recurse on each branch that is not already pure ($Info = 0$), using only that branch's rows and only the attributes not yet used on its path.

The table idiom, used unchanged in every problem below

Attribute	Value	Class 1	Class 2	$ D_j $	$Info(D_j)$	$\frac{ D_j }{ D } Info(D_j)$
-----------	-------	---------	---------	---------	-------------	-------------------------------

Time-savers: a pure branch contributes 0; a 1:1 branch has $Info$ exactly 1; branches with the same class ratio share one $Info$, so compute it once. Keep 4 decimals throughout.

Every entropy this section needs

$\log_2 x = \frac{\ln x}{\ln 2}$. Ratios are class counts in any order: 2:1 covers (2 Yes, 1 No) and (1 Yes, 2 No) alike. $Info = 0$ whenever only one class is present.

Split	1:1	2:1	3:1	3:2	4:1	5:1	4:3	6:1	9:5
$Info$	1.0000	0.9183	0.8113	0.9710	0.7219	0.6500	0.9852	0.5917	0.9403
Split	2:2:1	3:1:1	3:2:1	3:2:2	4:2:1	4:3:3	6:3:2	6:5:3	
$Info$	1.5219	1.3710	1.4591	1.5567	1.3788	1.5710	1.4354	1.5306	

Problem 1.1 10

QUESTION AS PRINTED 2079 Bhadra · Q3 · [10]

Draw decision tree for the given data using ID3 algorithm.

Age	Income	Student	Credit_Rating	Buy's_Computer
Youth	High	No	Fair	No
Youth	High	No	Excellent	No
Middle_Aged	High	No	Fair	Yes
Senior	Medium	No	Fair	Yes
Senior	Low	Yes	Fair	Yes
Senior	Low	Yes	Excellent	No
Middle_Aged	Low	Yes	Excellent	Yes
Youth	Medium	No	Fair	No
Youth	Low	Yes	Fair	Yes
Senior	Medium	Yes	Fair	Yes
Youth	Medium	Yes	Excellent	Yes
Middle_Aged	Medium	No	Excellent	Yes
Middle_Aged	High	Yes	Fair	Yes
Senior	Medium	No	Excellent	No

Step 1 Entropy of the whole set

14 rows: 9 Yes, 5 No.

$$Info(D) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.4098 + 0.5305 = \mathbf{0.9403}$$

Steps 2 and 3 Split on each attribute in turn

Attribute	Value	Yes	No	$ D_j $	$Info(D_j)$	$\frac{ D_j }{ D } Info(D_j)$
Age	Youth	2	3	5	0.9710	$\frac{5}{14}(0.9710) = 0.3468$
	Middle_Aged	4	0	4	0	0
	Senior	3	2	5	0.9710	$\frac{5}{14}(0.9710) = 0.3468$
$Info_{Age}(D) = 0.6936$						Gain = 0.9403 - 0.6936 = 0.2467
Income	High	2	2	4	1.0000	$\frac{4}{14}(1.0000) = 0.2857$
	Medium	4	2	6	0.9183	$\frac{6}{14}(0.9183) = 0.3936$
	Low	3	1	4	0.8113	$\frac{4}{14}(0.8113) = 0.2318$
$Info_{Income}(D) = 0.9111$						Gain = 0.9403 - 0.9111 = 0.0292
Student	No	3	4	7	0.9852	$\frac{7}{14}(0.9852) = 0.4926$
	Yes	6	1	7	0.5917	$\frac{7}{14}(0.5917) = 0.2959$
$Info_{Student}(D) = 0.7885$						Gain = 0.9403 - 0.7885 = 0.1518
Credit_Rating	Fair	6	2	8	0.8113	$\frac{8}{14}(0.8113) = 0.4636$
	Excellent	3	3	6	1.0000	$\frac{6}{14}(1.0000) = 0.4286$
$Info_{Credit_Rating}(D) = 0.8922$						Gain = 0.9403 - 0.8922 = 0.0481

Step 4 Compare

Attribute	Gain
Age	0.2467
Student	0.1518
Credit_Rating	0.0481
Income	0.0292

Answer. The root attribute is **Age**, with the highest information gain, **0.2467**.

Why Age wins

- Age is the only attribute with a **pure branch**: all four *Middle_Aged* rows (3, 7, 12, 13) are Yes, so it removes more uncertainty than any other.
- Income removes almost none: its three branches are nearly as mixed as the parent.
- Middle_Aged* becomes a **leaf** at once. *Youth* (2:3) and *Senior* (3:2) still have $Info = 0.9710$, so each needs its own split: Step 5.

Step 5 Recurse: Steps 1 to 4 again, inside each impure branch

Rows numbered 1 to 14 top to bottom. Each branch uses only its own rows and only the attributes Age has not used.

Age = Youth rows 1, 2, 8, 9, 11 2 Yes, 3 No $Info = 0.9710$

Attribute	Value (rows)	Yes	No	Info	Gain
Student	No (1,2,8)	0	3	0	
	Yes (9,11)	2	0	0	0.9710
Income	High (1,2)	0	2	0	
	Medium (8,11)	1	1	1.0000	
	Low (9)	1	0	0	0.5710
Credit_Rating	Fair (1,8,9)	1	2	0.9183	
	Excellent (2,11)	1	1	1.0000	0.0200

Student: residual 0, gain $0.9710 - 0.0000 = 0.9710$. Income: residual $\frac{2}{5}(1.0000) = 0.4000$, gain $0.9710 - 0.4000 = 0.5710$. Credit_Rating: residual $\frac{3}{5}(0.9183) + \frac{2}{5}(1.0000) = 0.9510$, gain $0.9710 - 0.9510 = 0.0200$. **Split on Student.**

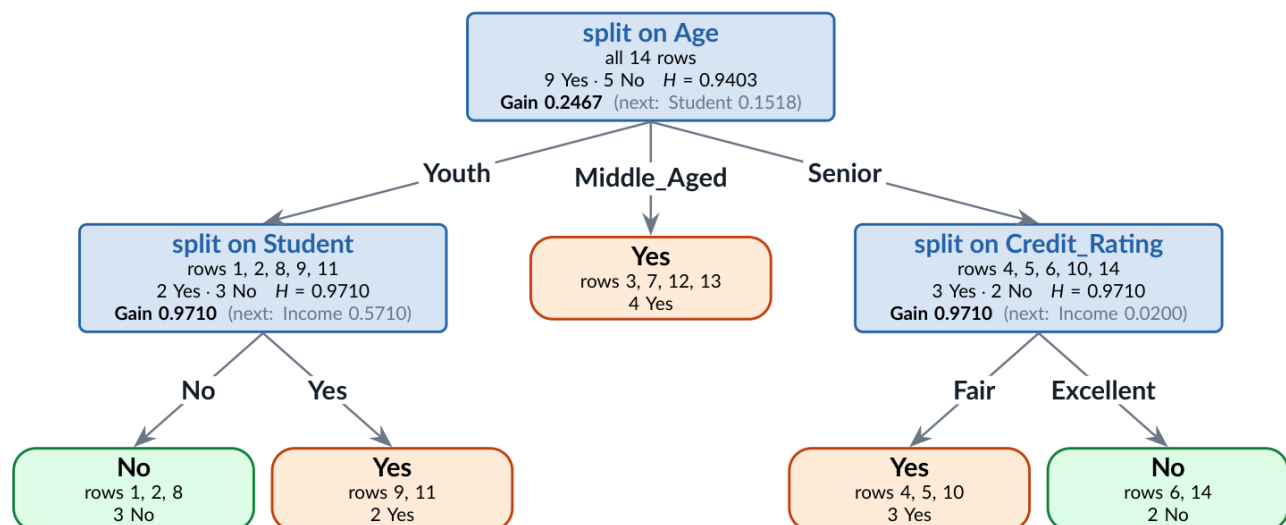
Age = Senior rows 4, 5, 6, 10, 14 3 Yes, 2 No $Info = 0.9710$

Attribute	Value (rows)	Yes	No	Info	Gain
Credit_Rating	Fair (4,5,10)	3	0	0	
	Excellent (6,14)	0	2	0	0.9710
Income	Medium (4,10,14)	2	1	0.9183	
	Low (5,6)	1	1	1.0000	0.0200
Student	No (4,14)	1	1	1.0000	
	Yes (5,6,10)	2	1	0.9183	0.0200

Credit_Rating: residual 0, gain $0.9710 - 0.0000 = 0.9710$. Income: residual $\frac{3}{5}(0.9183) + \frac{2}{5}(1.0000) = 0.9510$, gain $0.9710 - 0.9510 = 0.0200$. Student: residual $\frac{2}{5}(1.0000) + \frac{3}{5}(0.9183) = 0.9510$, gain $0.9710 - 0.9510 = 0.0200$. **Split on Credit_Rating.**

The tree, 2 levels of tests and 5 leaves, one rule per leaf: Age = Youth, Student = No $\rightarrow$ No; Age = Youth, Student = Yes $\rightarrow$ Yes; Age = Middle_Aged $\rightarrow$ Yes; Age = Senior, Credit_Rating = Fair $\rightarrow$ Yes; Age = Senior, Credit_Rating = Excellent $\rightarrow$ No. Income never appears: **ID3 has decided it is irrelevant**, which is the feature-selection side of the algorithm.

The finished tree: buys_computer (rows numbered 1 to 14 top to bottom)



5 leaves, depth 2; classifies all 14 training tuples correctly. The same tree as Han & Kamber 3rd ed Fig 8.2, whose Example 8.1 prints the same gains (to its own truncation: 0.246, 0.151).

Problem 1.2 8

QUESTION AS PRINTED 2078 Bhadra · Q4 · [8]

Construct a decision tree for the following data set using information gain.

Predict the class label for a data point with values <Female, 2, standard, high>

Gender	Car ownership	Travel cost	Income level	Transport mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Female	1	Cheap	Medium	Train
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train

Step 1 Entropy of the whole set three classes, so three terms

10 rows: 4 Bus, 3 Car, 3 Train.

$$Info(D) = -\frac{4}{10} \log_2 \frac{4}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{3}{10} \log_2 \frac{3}{10} = 0.5288 + 0.5211 + 0.5211 = \mathbf{1.5710}$$

Steps 2 and 3 Split on each attribute in turn class columns (Bus, Car, Train)

Attribute	Value	Bus	Car	Train	$ D_j $	$Info(D_j)$	$\frac{ D_j }{10} Info(D_j)$
Gender	Male	3	1	1	5	1.3710	$\frac{5}{10}(1.3710) = 0.6855$
	Female	1	2	2	5	1.5219	$\frac{5}{10}(1.5219) = 0.7610$
$Info_{Gender}(D) = 1.4465$							Gain = $1.5710 - 1.4465 = 0.1245$
Car ownership	0	2	0	1	3	0.9183	$\frac{3}{10}(0.9183) = 0.2755$
	1	2	1	2	5	1.5219	$\frac{5}{10}(1.5219) = 0.7610$
	2	0	2	0	2	0	0
$Info_{Car\ ownership}(D) = 1.0365$							Gain = $1.5710 - 1.0365 = 0.5345$
Travel cost	Cheap	4	0	1	5	0.7219	$\frac{5}{10}(0.7219) = 0.3609$
	Expensive	0	3	0	3	0	0
	Standard	0	0	2	2	0	0
$Info_{Travel\ cost}(D) = 0.3609$							Gain = $1.5710 - 0.3609 = \mathbf{1.2101}$
Income level	Low	2	0	0	2	0	0
	Medium	2	1	3	6	1.4591	$\frac{6}{10}(1.4591) = 0.8755$
	High	0	2	0	2	0	0
$Info_{Income\ level}(D) = 0.8755$							Gain = $1.5710 - 0.8755 = 0.6955$

Step 4 Compare

Attribute	Gain
Travel cost	1.2101
Income level	0.6955
Car ownership	0.5345
Gender	0.1245

Why Travel cost wins

- Two of its three branches are already **pure**: Expensive is all Car, Standard all Train. No other attribute leaves so little mixed.
- Gender barely helps: Male (3,1,1) and Female (1,2,2) are each almost as mixed as the whole set.
- Only the Cheap branch needs splitting** (4 Bus, 1 Train): Step 5.

Answer. The root attribute is **Travel cost**, with the highest information gain, **1.2101**.

Step 5 Recurse on the Cheap branch, then on what it leaves mixed

Rows numbered 1 to 10 top to bottom. Only Bus and Train occur in this branch, so the Car column is dropped.

Travel cost = Cheap rows 1, 2, 3, 4, 8 4 Bus, 1 Train

$Info = 0.7219$

Attribute	Value (rows)	Bus	Train	Info	Gain
Gender	Male (1,2,4)	3	0	0	0.3219
	Female (3,8)	1	1	1.0000	
Car ownership	0 (1,3)	2	0	0	0.1709
	1 (2,4,8)	2	1	0.9183	
Income level	Low (1,3)	2	0	0	0.1709
	Medium (2,4,8)	2	1	0.9183	

Gender: residual $\frac{2}{5}(1.0000) = 0.4000$, gain $0.7219 - 0.4000 = 0.3219$.

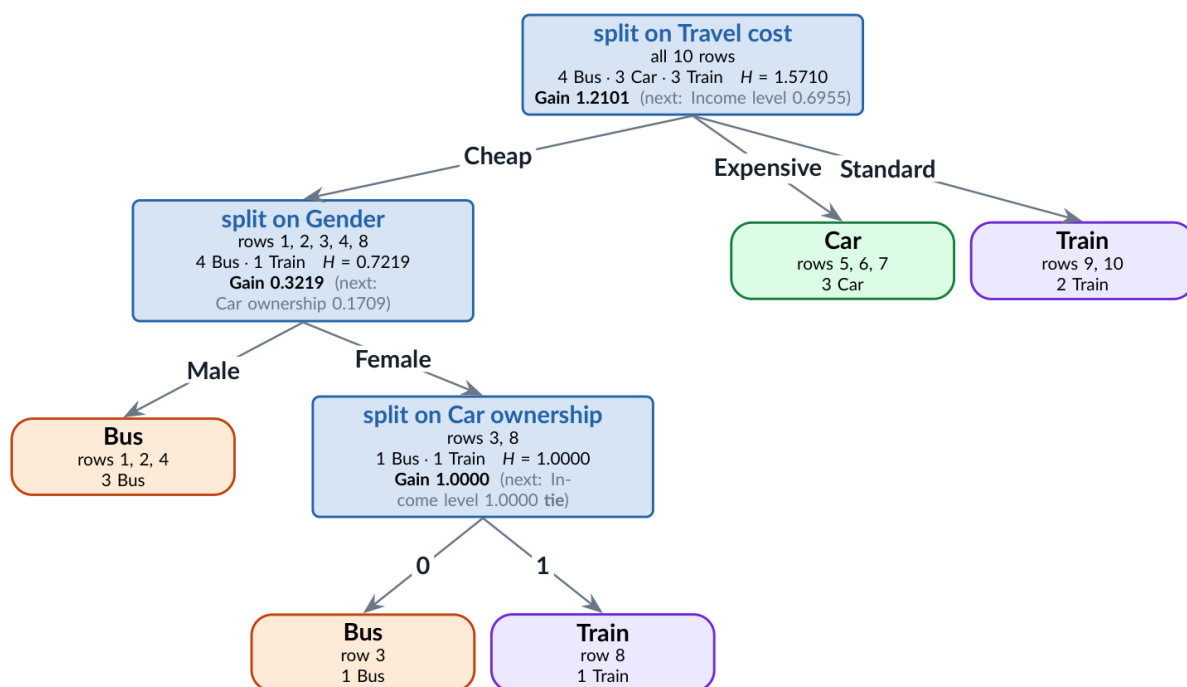
Car ownership: residual $\frac{3}{5}(0.9183) = 0.5510$, gain

$0.7219 - 0.5510 = 0.1709$. Income level: residual $\frac{3}{5}(0.9183) = 0.5510$,

gain $0.7219 - 0.5510 = 0.1709$. **Split on Gender.**

The tree, 3 levels of tests and 5 leaves, one rule per leaf: Travel cost = Cheap, Gender = Male $\rightarrow$ Bus; Travel cost = Cheap, Gender = Female, Car ownership = 0 $\rightarrow$ Bus; Travel cost = Cheap, Gender = Female, Car ownership = 1 $\rightarrow$ Train; Travel cost = Expensive $\rightarrow$ Car; Travel cost = Standard $\rightarrow$ Train. Income level never appears: **ID3 has decided it is irrelevant**, which is the feature-selection side of the algorithm.

The finished tree: transport mode (rows numbered 1 to 10 top to bottom)



5 leaves, depth 3; classifies all 10 training tuples correctly.

Prediction for <Female, 2, standard, high>

- Read the root first: Travel cost = **standard**.
- That branch is **pure Train**, so the walk stops immediately.
- **No other attribute is consulted** – Gender, car ownership and income are never reached. **Answer: Train.**
- **This is the point of the question:** a decision tree only tests the attributes on the path it takes, not all of them.

Problem 1.3 2+9+3

QUESTION AS PRINTED 2082 Bhadra · Q3 · [2+9+3]

When do we use decision tree? Construct a complete decision tree using ID-3 algorithm for following dataset. How do you evaluate the performance of this model?

SN	Credit history	Debt	Collateral	Income	Credit Risk
1	bad	high	none	\$0 to \$15K	high
2	unknown	high	none	\$15 to \$35K	high
3	unknown	low	none	\$15 to \$35K	moderate
4	unknown	low	none	\$0 to \$15K	high
5	unknown	low	none	over \$35K	low
6	unknown	low	adequate	over \$35K	low
7	bad	low	none	\$0 to \$15K	high
8	bad	low	adequate	over \$35K	moderate
9	good	low	none	over \$35K	low
10	good	high	adequate	over \$35K	low
11	good	high	none	\$0 to \$15K	high
12	good	high	none	\$15 to \$35K	moderate
13	good	high	none	over \$35K	low
14	bad	high	none	\$15 to \$35K	high

Three classes, not two, so *Info* has three terms. Rows are the table's SN.

Step 1 Entropy of the whole set

14 rows: 6 high, 3 moderate, 5 low.

$$Info(D) = -\frac{6}{14} \log_2 \frac{6}{14} - \frac{3}{14} \log_2 \frac{3}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.5239 + 0.4762 + 0.5305 = \mathbf{1.5306}$$

Steps 2 and 3 Split on each attribute in turn class columns (high, moderate, low)

Attribute	Value	high	mod	low	$ D_j $	$Info(D_j)$	$\frac{ D_j }{14} Info(D_j)$
Credit history	bad	3	1	0	4	0.8113	$\frac{4}{14}(0.8113) = 0.2318$
	unknown	2	1	2	5	1.5219	$\frac{5}{14}(1.5219) = 0.5435$
	good	1	1	3	5	1.3710	$\frac{5}{14}(1.3710) = 0.4896$
$Info_{Credit\ history}(D) = 1.2649$							Gain = 1.5306 – 1.2649 = 0.2657
Debt	high	4	1	2	7	1.3788	$\frac{7}{14}(1.3788) = 0.6894$
	low	2	2	3	7	1.5567	$\frac{7}{14}(1.5567) = 0.7783$
$Info_{Debt}(D) = 1.4677$							Gain = 1.5306 – 1.4677 = 0.0629
Collateral	none	6	2	3	11	1.4354	$\frac{11}{14}(1.4354) = 1.1278$
	adequate	0	1	2	3	0.9183	$\frac{3}{14}(0.9183) = 0.1968$
$Info_{Collateral}(D) = 1.3246$							Gain = 1.5306 – 1.3246 = 0.2060
Income	\$0–15K	4	0	0	4	0	0
	\$15–35K	2	2	0	4	1.0000	$\frac{4}{14}(1.0000) = 0.2857$
	over \$35K	0	1	5	6	0.6500	$\frac{6}{14}(0.6500) = 0.2786$
$Info_{Income}(D) = 0.5643$							Gain = 1.5306 – 0.5643 = 0.9663

Step 4 Compare

Attribute	Gain
Income	0.9663
Credit history	0.2657
Collateral	0.2060
Debt	0.0629

Why Income wins, by a wide margin

- Its \$0–15K branch is **pure**: rows 1, 4, 7, 11 are all high risk.
- Its over \$35K branch is nearly pure (1 moderate, 5 low), $Info = 0.6500$.
- Debt splits 7/7 into two branches about as mixed as the parent, so it gains almost nothing.

Answer. The root attribute is **Income**, with the highest information gain, **0.9663**.

Step 5 Recurse on the two impure income bands

Only the classes present in a branch get a column.

Income = \$15–35K rows 2, 3, 12, 14 2 high, 2 moderate

$Info = 1.0000$

Attribute	Value (rows)	high	mod	Info	Gain
Credit history	unknown (2,3)	1	1	1.0000	
	good (12)	0	1	0	
	bad (14)	1	0	0	0.5000
Debt	high (2,12,14)	2	1	0.9183	
	low (3)	0	1	0	0.3113
Collateral	none (2,3,12,14)	2	2	1.0000	0.0000

Credit history: residual $\frac{2}{4}(1.0000) = 0.5000$, gain

$1.0000 - 0.5000 = 0.5000$. Debt: residual $\frac{3}{4}(0.9183) = 0.6887$, gain

$1.0000 - 0.6887 = 0.3113$. Collateral: residual 1.0000, gain

$1.0000 - 1.0000 = 0.0000$. **Split on Credit history.**

Income = \$15–35K, Credit history = unknown rows 2, 3 1

high, 1 moderate $Info = 1.0000$

Attribute	Value (rows)	high	mod	Info	Gain
Debt	high (2)	1	0	0	
	low (3)	0	1	0	1.0000
Collateral	none (2,3)	1	1	1.0000	0.0000

Debt: residual 0, gain $1.0000 - 0.0000 = 1.0000$. Collateral: residual

1.0000 , gain $1.0000 - 1.0000 = 0.0000$. **Split on Debt.**

Income = over \$35K rows 5, 6, 8, 9, 10, 13 1 moderate, 5 low

$Info = 0.6500$

Attribute	Value (rows)	mod	low	Info	Gain
Credit history	unknown (5,6)	0	2	0	
	bad (8)	1	0	0	
	good (9,10,13)	0	3	0	0.6500
Collateral	none (5,9,13)	0	3	0	
Debt	adequate (6,8,10)	1	2	0.9183	0.1908
	low (5,6,8,9)	1	3	0.8113	
	high (10,13)	0	2	0	0.1091

Credit history: residual 0, gain $0.6500 - 0.0000 = 0.6500$. Collateral:

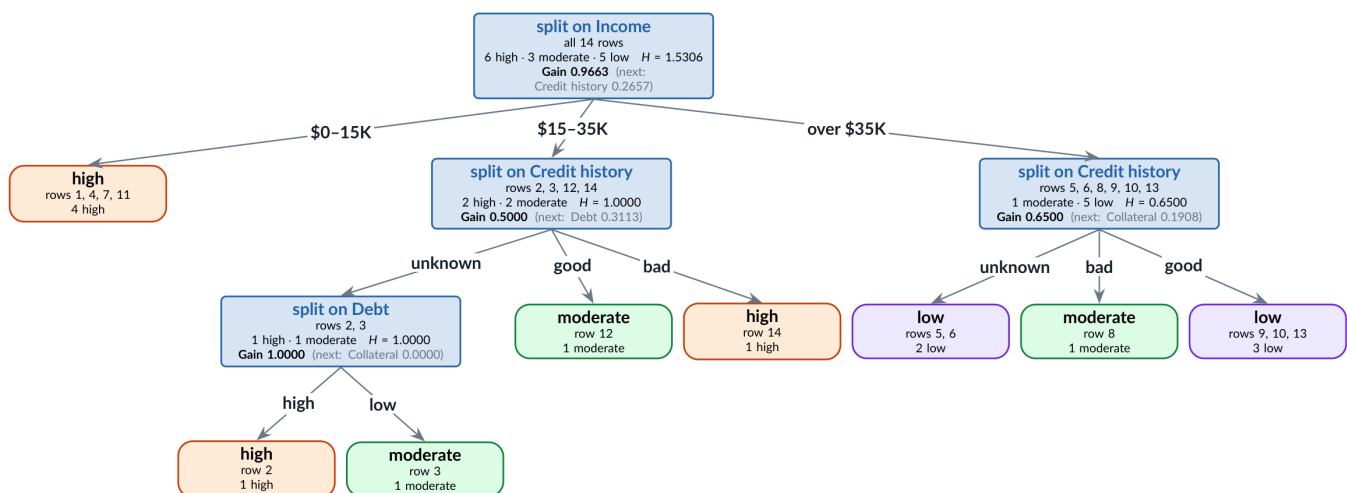
residual $\frac{3}{6}(0.9183) = 0.4592$, gain $0.6500 - 0.4592 = 0.1908$. Debt:

residual $\frac{4}{6}(0.8113) = 0.5409$, gain $0.6500 - 0.5409 = 0.1091$. **Split on**

Credit history.

The tree, 3 levels of tests and 8 leaves, one rule per leaf: Income = \$0–15K → high; Income = \$15–35K, Credit history = unknown, Debt = high → high; Income = \$15–35K, Credit history = unknown, Debt = low → moderate; Income = \$15–35K, Credit history = good → moderate; Income = \$15–35K, Credit history = bad → high; Income = over \$35K, Credit history = unknown → low; Income = over \$35K, Credit history = bad → moderate; Income = over \$35K, Credit history = good → low. Collateral never appears: **ID3 has decided it is irrelevant**, which is the feature-selection side of the algorithm.

The finished tree: credit risk (rows are the table's SN)



Reading the tree

- 8 leaves; classifies all 14 tuples correctly.
- **Income** alone decides the \$0–15K band (all high risk).
- **Credit history** is tested twice, once under each other income band: an attribute used on one path can be reused on another.
- Only the \$15–35K, unknown-history tuples need a third test, **Debt**.

The 3-mark part: evaluating the model

- **Do not evaluate on the training set** – this tree scores 100 % there by construction, which measures nothing.
- Hold out data, or better **stratified 10-fold cross-validation** (§3.7).
- Build a **confusion matrix** on the held-out tuples, then report **accuracy**, and per class **precision**, **recall** and F_1 . With three classes, report them **per class** – a single accuracy hides which risk band is being missed.
- Mention **pruning** and a validation set, since a fully grown ID3 tree overfits.

Problem 1.4 – root node only 2+8

QUESTION AS PRINTED 2081 Bhadra · Q5 · [2+8]

State an algorithm for constructing a decision tree. Using ID3 identify the root node for the following data set.

Gender	Car Ownership	Travel Cost	Income Level	Transport Mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	High	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car

A different 10-row table from Problem 1.2, despite looking almost identical: four rows differ. **Only the root is asked for**, so stop after Step 4; do not grow the whole tree.

Step 1 Entropy of the whole set

10 rows: 4 Bus, 3 Train, 3 Car.

$$Info(D) = -\frac{4}{10} \log_2 \frac{4}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{3}{10} \log_2 \frac{3}{10} = 0.5288 + 0.5211 + 0.5211 = \mathbf{1.5710}$$

Steps 2 and 3 Split on each attribute in turn class columns (Bus, Train, Car)

Attribute	Value	Bus	Train	Car	$ D_j $	$Info(D_j)$	$\frac{ D_j }{10} Info(D_j)$
Gender	Male	3	1	1	5	1.3710	$\frac{5}{10}(1.3710) = 0.6855$
	Female	1	2	2	5	1.5219	$\frac{5}{10}(1.5219) = 0.7610$
$Info_{Gender}(D) = 1.4465$							Gain = $1.5710 - 1.4465 = 0.1245$
Car ownership	0	2	1	0	3	0.9183	$\frac{3}{10}(0.9183) = 0.2755$
	1	2	2	1	5	1.5219	$\frac{5}{10}(1.5219) = 0.7610$
	2	0	0	2	2	0	0
$Info_{Car ownership}(D) = 1.0365$							Gain = $1.5710 - 1.0365 = 0.5345$
Travel cost	Cheap	4	1	0	5	0.7219	$\frac{5}{10}(0.7219) = 0.3609$
	Standard	0	2	0	2	0	0
	Expensive	0	0	3	3	0	0
$Info_{Travel cost}(D) = 0.3609$							Gain = $1.5710 - 0.3609 = 1.2101$
Income level	Low	2	0	0	2	0	0
	Medium	2	2	1	5	1.5219	$\frac{5}{10}(1.5219) = 0.7610$
	High	0	1	2	3	0.9183	$\frac{3}{10}(0.9183) = 0.2755$
$Info_{Income level}(D) = 1.0365$							Gain = $1.5710 - 1.0365 = 0.5345$

Step 4 Compare

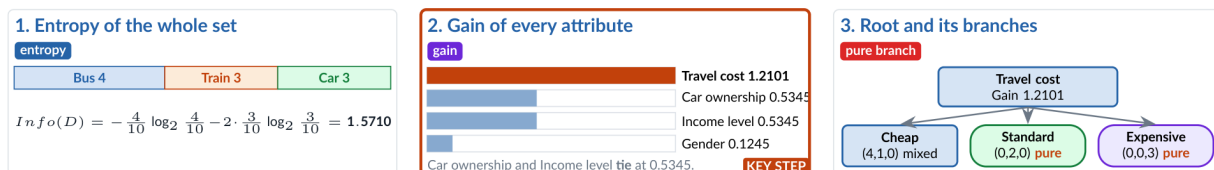
Attribute	Gain
Travel cost	1.2101
Car ownership	0.5345 tie with Income level
Income level	0.5345 tie with Car ownership
Gender	0.1245

Answer. The root attribute is **Travel cost**, with the highest information gain, **1.2101**.

Why Travel cost wins

- Two of its three branches are already **pure** (Standard all Train, Expensive all Car), so it removes most of the uncertainty in one test. **Say that**; do not just quote the number.
- Its gain is more than double the next best.
- Car ownership and Income level tie exactly.** Note it: had the question asked for the second level, you would have to break that tie explicitly.

Step by step: the root node, and why it wins



M2. Gini index

Method

$$Gini(D) = 1 - \sum_i p_i^2 \quad Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2) \quad \Delta Gini = Gini(D) - Gini_A(D)$$

- Gini uses BINARY splits only.** For an attribute with v values you must try all $2^{v-1} - 1$ ways of splitting the values into 2 subsets and keep the best.
- Pick the largest $\Delta Gini$ (equivalently the smallest $Gini_A$).

Worked example (same 14-tuple dataset) 6

QUESTION AS PRINTED 2075 Chaitra · Q3 · [6+4]

Define Decision Tree Classifier with Gini-Index with suitable example. How can you handle overfitting in Decision Tree?

QUESTION AS PRINTED 2071 Chaitra · Q3 · [10]

What is a decision tree? Explain Gini Index with suitable example.

Neither paper prints a dataset – the “suitable example” is yours to supply. The 14-tuple *buys_computer* set below is the safe one to reuse. verified/added

- $Gini(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 1 - 0.4133 - 0.1276 = \mathbf{0.459}$
- Split **Income** as {low, medium} vs {high}:
 - $D_1 = 10$ tuples (7 Yes, 3 No): $Gini = 1 - (0.7)^2 - (0.3)^2 = 0.420$
 - $D_2 = 4$ tuples (2 Yes, 2 No): $Gini = 1 - (0.5)^2 - (0.5)^2 = 0.500$
 - $Gini_{Income}(D) = \frac{10}{14}(0.420) + \frac{4}{14}(0.500) = 0.300 + 0.143 = 0.443$
 - $\Delta Gini = 0.459 - 0.443 = \mathbf{0.016}$
- Split **Age** as {youth, senior} vs {middle_aged}:
 - $D_1 = 10$ tuples (5 Yes, 5 No): $Gini = 0.500$
 - $D_2 = 4$ tuples (4 Yes, 0 No): $Gini = \mathbf{0}$ (pure)
 - $Gini_{Age}(D) = \frac{10}{14}(0.500) + \frac{4}{14}(0) = 0.357$
 - $\Delta Gini = 0.459 - 0.357 = \mathbf{0.102} \rightarrow \text{best, so Age is chosen}$
- Same winner as information gain, which is the usual (not guaranteed) outcome.

Step by step: Gini on two candidate binary splits

1. Gini of the whole set

Gini

Yes 9	No 5
-------	------

$$1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 0.459$$

2. Income

binary split

{low, medium}: 10 tuples

7	3	0.420
---	---	-------

{high}: 4 tuples

2	2	0.500
---	---	-------

$$Gini_A = 0.443, \Delta = 0.016$$

3. Age

binary split pure branch

{youth, senior}: 10 tuples

5	5	0.500
---	---	-------

{middle_aged}: 4 tuples

4	0 pure
---	--------

$$Gini_A = 0.357, \Delta = 0.102$$

KEY STEP

4. Largest drop wins

largest wins

Income Δ 0.016Age Δ 0.102

Age is chosen, as by information gain.

M3. Naive Bayes classification

Method

1. Priors: $P(C_i) = |C_{i,D}|/|D|$.
2. For each attribute value in X : $P(x_k|C_i)$ from counts within class C_i .
3. $P(X|C_i) = \prod_k P(x_k|C_i)$.
4. Compare $P(X|C_i)P(C_i)$ across classes; **largest wins**.
 - **Never divide by $P(X)$** – it is identical for every class and cancels.
 - Any zero count $\rightarrow$ apply **Laplacian correction** $\frac{n_k+1}{|C_{i,D}|+k}$.
 - Lay the work out as a small table of conditional probabilities; it earns method marks even if arithmetic slips.

Problem 3.1 10

QUESTION AS PRINTED 2072 Chaitra · Q3 · [10]

Predict Class label using naive Bayesian classifier for $X = (\text{age} = \text{youth}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit-rating} = \text{fair})$ using the following data set.

RID	Age	Income	Student	Credit-rating	Class Buy computer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle-age	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle-age	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Middle-age	Medium	No	Excellent	Yes
13	Middle-age	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No

Given

14 tuples. $P(\text{Yes}) = 9/14 = 0.643$, $P(\text{No}) = 5/14 = 0.357$.

Conditional probabilities

	Yes	No
$P(\text{age} = \text{youth})$	$2/9 = 0.222$	$3/5 = 0.600$
$P(\text{income} = \text{med})$	$4/9 = 0.444$	$2/5 = 0.400$
$P(\text{student} = \text{yes})$	$6/9 = 0.667$	$1/5 = 0.200$
$P(\text{credit} = \text{fair})$	$6/9 = 0.667$	$2/5 = 0.400$

Combine

- $P(X|\text{Yes}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = \mathbf{0.0439}$
- $P(X|\text{No}) = 0.600 \times 0.400 \times 0.200 \times 0.400 = \mathbf{0.0192}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.0439 \times 0.643 = \mathbf{0.028}$
- $P(X|\text{No})P(\text{No}) = 0.0192 \times 0.357 = \mathbf{0.007}$

Final answer

$0.028 > 0.007 \rightarrow \text{buys_computer} = \text{Yes}.$

- Matches Han & Kamber 3rd ed Example 8.4.

Step by step: Naive Bayes for $X = (\text{youth, medium, yes, fair})$

1. Priors

prior

Yes 9 No 5

$$P(\text{Yes}) = 9/14 = 0.643$$

$$P(\text{No}) = 5/14 = 0.357$$

2. Count within each class

class count

	Yes	No
age = youth	2/9	3/5
income = med	4/9	2/5
student = yes	6/9	1/5
credit = fair	6/9	2/5

3. Multiply

product

$$P(X|\text{Yes}) = 0.222 \cdot 0.444 \cdot 0.667 \cdot 0.667$$

$$= 0.0439$$

$$P(X|\text{No}) = 0.600 \cdot 0.400 \cdot 0.200 \cdot 0.400$$

$$= 0.0192$$

Attributes treated as independent given the class.

4. Times the prior, compare

largest wins

Yes: 0.028

No: 0.007

 $\times P(C)$, then compare. $P(X)$ cancels.

buys_computer = Yes

KEY STEP

Problem 3.2 6

QUESTION AS PRINTED 2079 Bhadra · Q4 · [6]

Suppose you have a test record " $X = (\text{Home Owner} = \text{No}, \text{Marital Status} = \text{Married}, \text{Income} = \$120\text{K})$ ". Your job is to classify this record using Naive Bayesian Classification. Use the following table for your calculations.

Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given

Defaulted: **No** = 7, **Yes** = 3. $P(\text{No}) = 0.7$, $P(\text{Yes}) = 0.3$.

Step 1: categorical attributes

- $P(\text{HO} = \text{No} | \text{No}) = 4/7 = 0.571$ $P(\text{HO} = \text{No} | \text{Yes}) = 3/3 = 1.0$
- $P(\text{Married} | \text{No}) = 4/7 = 0.571$ $P(\text{Married} | \text{Yes}) = 0/3 = 0 \leftarrow \text{zero!}$
- Apply **Laplacian correction** (Marital Status has $k = 3$ values): $P(\text{Married} | \text{Yes}) = \frac{0+1}{3+3} = 0.167$; and $P(\text{HO} = \text{No} | \text{Yes}) = \frac{3+1}{3+2} = 0.8$.

Step 2: continuous attribute (Income), Gaussian

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Class **No**: incomes 125, 100, 70, 120, 60, 220, 75 $\rightarrow \mu = 110, \sigma = 54.54$

$$g(120) = \frac{1}{\sqrt{2\pi}(54.54)} e^{-\frac{(10)^2}{2(2975)}} = 0.00719$$

- Class **Yes**: incomes 95, 85, 90 $\rightarrow \mu = 90, \sigma = 5$

$$g(120) = \frac{1}{\sqrt{2\pi}(5)} e^{-\frac{(30)^2}{2(25)}} = 0.0798 e^{-18} \approx 1.2 \times 10^{-9}$$

- **The Yes class has a tiny spread ($\sigma = 5$) around \$90K, so \$120K is 6 standard deviations out** — effectively impossible under that model.

Step 3: combine

- $P(X|\text{No})P(\text{No}) = 0.571 \times 0.571 \times 0.00719 \times 0.7 = 1.64 \times 10^{-3}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.8 \times 0.167 \times 1.2 \times 10^{-9} \times 0.3 \approx 4.8 \times 10^{-11}$

Final answer

Defaulted Borrower = No (larger by roughly 10^7).

Step by step: a zero count, the Laplace fix and a Gaussian attribute

1. Priors

prior

No 7	Yes 3
------	-------

$P(\text{No}) = 0.7$
 $P(\text{Yes}) = 0.3$

2. Categorical counts

class count

HO = No	No 4/7	Yes 3/3
Married	No 4/7	Yes 0/3

A single zero makes $P(X|\text{Yes}) = 0$ whatever else holds.

3. Laplace fix

Laplace

$P(\text{Married}|\text{Yes}) = \frac{0+1}{3+3} = 0.167$
 $P(\text{HO}=\text{No}|\text{Yes}) = \frac{3+1}{3+3} = 0.8$

+1 per count, +k per total (k = values the attribute takes: 3, 2).

4. Income: a Gaussian per class

Gaussian

5. Combine, compare

product largest wins

No: $0.571 \cdot 0.571 \cdot 0.00719 \cdot 0.7$
 $= 1.64 \times 10^{-3}$
 Yes: $0.8 \cdot 0.167 \cdot 1.2 \times 10^{-9} \cdot 0.3$
 $\approx 4.8 \times 10^{-11}$
 Defaulted = No, larger by about 10^7 .

M4. k -Nearest Neighbour

Method

1. Compute the Euclidean distance from the query to **every** training tuple.
 2. Sort ascending, take the k smallest.
 3. **Majority class** among those k is the prediction.
- **Show the full distance table** – that is where the marks are, not the final label.
 - You may compare squared distances to save time, but write the roots in the final table.

Problem 4.1 6

QUESTION AS PRINTED 2080 Bhadra · Q2 (b) · [3+6]

When do we use classifier? You have the following information about the flower. Your job is to classify the given flower with SepalLength = 7, SepalWidth = 3.2, PetalLength = 4.7, and PetalWidth = 1.4. Use KNN algorithm for $K = 3$ with Euclidean distance matrix.

Id	Sepal_Lenth	Spal_Width	Petal_Lenth	Petal_Width	Labal
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	6	2.2	4	1	versicolor
5	6.1	2.9	4.7	1.4	versicolor
6	5.6	2.9	3.6	1.3	versicolor
7	6.7	3.1	4.4	1.4	versicolor

Column names are

the paper's own spellings.

Id	Tuple	Squared differences	d	Label
1	(5.1, 3.5, 1.4, 0.2)	$3.61 + 0.09 + 10.89 + 1.44 = 16.03$	4.004	setosa
2	(4.9, 3.0, 1.4, 0.2)	$4.41 + 0.04 + 10.89 + 1.44 = 16.78$	4.096	setosa
3	(4.7, 3.2, 1.3, 0.2)	$5.29 + 0 + 11.56 + 1.44 = 18.29$	4.277	setosa
4	(6.0, 2.2, 4.0, 1.0)	$1 + 1 + 0.49 + 0.16 = 2.65$	1.628	versicolor
5	(6.1, 2.9, 4.7, 1.4)	$0.81 + 0.09 + 0 + 0 = 0.90$	0.949	versicolor
6	(5.6, 2.9, 3.6, 1.3)	$1.96 + 0.09 + 1.21 + 0.01 = 3.27$	1.808	versicolor
7	(6.7, 3.1, 4.4, 1.4)	$0.09 + 0.01 + 0.09 + 0 = 0.19$	0.436	versicolor

Step: pick the 3 nearest

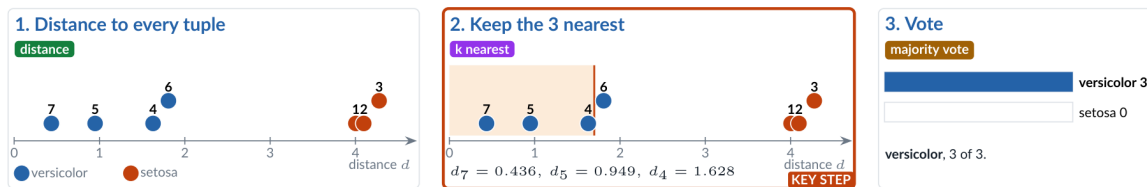
- Sorted: $d_7 = 0.436 < d_5 = 0.949 < d_4 = 1.628 < d_6 = 1.808 < d_1 = 4.004 \dots$
- The 3 nearest are **Id 7, Id 5, Id 4** → versicolor, versicolor, versicolor.

Final answer

versicolor (unanimous, 3 of 3).

- Note all 3 setosa tuples sit at distance > 4 : they have tiny petals, so they are nowhere near the query. The vote was never in doubt.

Step by step: 3-NN on the flower



M5. Confusion matrix metrics (8 papers — highest-yield numerical in the syllabus)

DO THIS FIRST, EVERY TIME. Copy the row and column headings onto your answer sheet and mark which axis is **actual**. The IOE papers **flip the orientation between years**.

Proof: 75 Ash and 72 Ch print the *identical* four numbers 21, 6, 7, 41 — but 75 Ash has actual on the rows and 72 Ch has predicted on the rows. The result is that **sensitivity and precision swap places** (0.778 / 0.750 becomes 0.750 / 0.778). Same numbers, different answers. **verified/added**

Method

- Identify TP , FN , FP , TN from the orientation. Then $P = TP + FN$ (all actual positives), $N = FP + TN$ (all actual negatives).
- Accuracy = $\frac{TP+TN}{P+N}$ Error = 1–accuracy
- Sensitivity = recall = $TPR = \frac{TP}{P}$ Specificity = $TNR = \frac{TN}{N}$
- FPR = false alarm rate = $\frac{FP}{N} = 1 - \text{specificity}$ Precision = $\frac{TP}{TP+FP}$

Problem 5.1 5

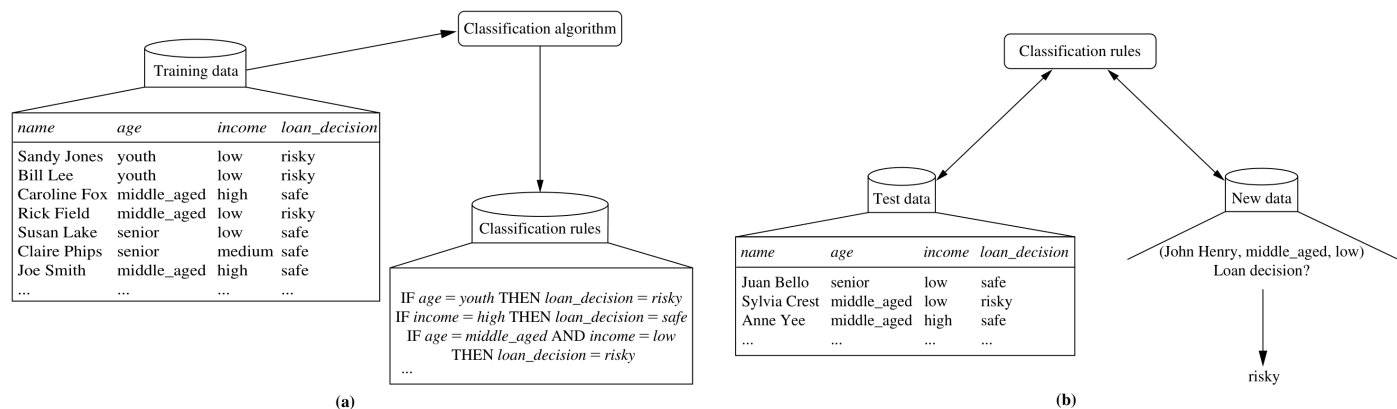
QUESTION AS PRINTED 2074 Chaitra · Q3 · [2+3+5]

Draw clear block diagram depicting different stages in classification. Explain the inverse relation between precision and recall. Given the confusion matrix, determine accuracy, sensitivity and precision of the classifier model.

Actual \ Predicted	Positive	Negative
Positive	142	40
Negative	98	720

“Positive” is the paper’s own spelling.

The [2] block diagram



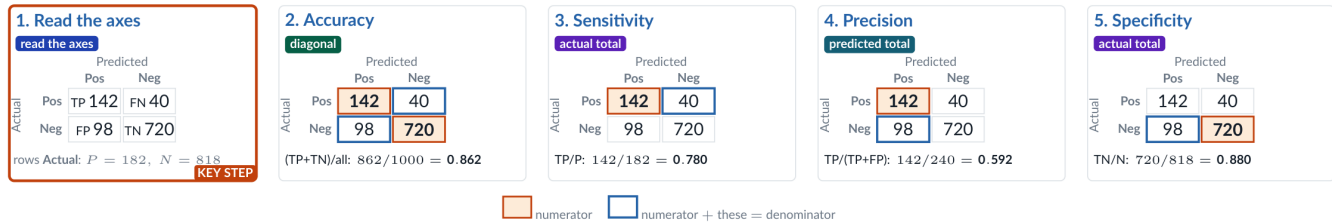
(a) learning: training data → algorithm → model; (b) classification: model checked on test data, then applied to new data. The precision–recall part is Ch 3 theory, §3.7. H&K Fig 8.1.

Step 1: read the four cells

rows are **Actual here**, so $TP = 142$, $FN = 40$, $FP = 98$, $TN = 720 \rightarrow P = 182$, $N = 818$, total = 1000.

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	$(142 + 720)/1000$	0.862
Sensitivity = TPR	TP/P	$142/182$	0.780
Precision	$TP/(TP + FP)$	$142/240$	0.592
Specificity	TN/N	$720/818$	0.880

Step by step: read the axes, then each metric's cells (rows Actual)



Problem 5.2 4

QUESTION AS PRINTED 2078 Bhadra · Q6 · [4]

Calculate: Accuracy, TPR, FPR and Precision for the given confusion matrix for a classifier.

		Actual Class	
		Class 1	Class 2
Predicted Class	Class 1	142	40
	Class 2	98	720

Same four numbers as 5.1, but the rows are Predicted here.

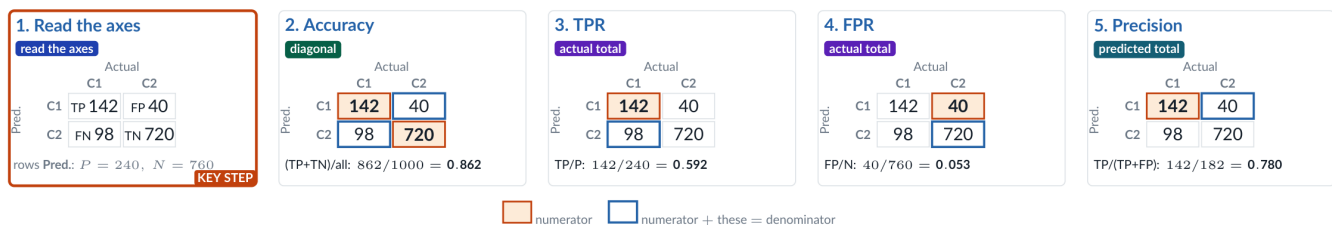
Step 1: read the four cells

rows are Predicted here, so $TP = 142$, $FP = 40$, $FN = 98$, $TN = 720 \rightarrow P = 142 + 98 = 240$, $N = 40 + 720 = 760$.

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	$(142 + 720)/1000$	0.862
TPR	TP/P	$142/240$	0.592
FPR	FP/N	$40/760$	0.053
Precision	$TP/(TP + FP)$	$142/182$	0.780

- Accuracy is **unchanged from 5.1** – the diagonal is the same either way round.
- **Compare with 5.1**: sensitivity and precision have exchanged values. This is the orientation trap, live.

Step by step: read the axes, then each metric's cells (rows Predicted)



Problems 5.3 and 5.4 10

QUESTION AS PRINTED 2075 Ashwin · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	21	6
	Class 2	7	41

Calculate: Accuracy, Sensitivity, Specificity and Precision.

QUESTION AS PRINTED 2072 Chaitra · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		actual class	
		class1	class2
predicted class	class1	21	6
	class2	7	41

Calculate a. accuracy b. sensitivity c. specificity d. precision e. recall

Same four numbers, opposite orientation – that is the whole point of solving them side by side.

Both at once, so the orientation trap is visible

- 75 Ash, rows are **Actual**: $TP = 21$, $FN = 6$, $FP = 7$, $TN = 41$; $P = 27$, $N = 48$.
- 72 Ch, rows are **Predicted**: $TP = 21$, $FP = 6$, $FN = 7$, $TN = 41$; $P = 28$, $N = 47$.
- Both total 75.

Metric	Formula	75 Ash (rows Actual)		72 Ch (rows Predicted)	
		Subst.	Value	Subst.	Value
Accuracy	$(TP+TN)/(P+N)$	62/75	0.827	62/75	0.827
Sensitivity = recall	TP/P	21/27	0.778	21/28	0.750
Specificity	TN/N	41/48	0.854	41/47	0.872
Precision	$TP/(TP+FP)$	21/28	0.750	21/27	0.778

- **Accuracy is identical, every other metric moves.** Sensitivity and precision simply swap. Read the row/column labels before touching the arithmetic.

Step by step: read the axes, then each metric's cells (the same four numbers, both ways round)

1. Read the axes
read the axes

75 Ash		Predicted	72 Ch	Actual			
		C1 C2		C1 C2			
Actual	C1	TP 21	FP 6	Pred.	C1	TP 21	FP 6
	C2	FN 7	TN 41		C2	FN 7	TN 41

rows Actual: $P = 27$, $N = 48$ rows Pred: $P = 28$, $N = 47$

KEY STEP

2. Accuracy
diagonal

75 Ash		Predicted	72 Ch	Actual			
		C1 C2		C1 C2			
Actual	C1	21	6	Pred.	C1	21	6
	C2	7	41		C2	7	41

$(TP+TN)/all: 62/75 = 0.827$ $62/75 = 0.827$

3. Sensitivity = recall
actual total

75 Ash		Predicted	72 Ch	Actual			
		C1 C2		C1 C2			
Actual	C1	21	6	Pred.	C1	21	6
	C2	7	41		C2	7	41

$TP/P: 21/27 = 0.778$ $21/28 = 0.750$

4. Specificity
actual total

75 Ash		Predicted	72 Ch	Actual			
		C1 C2		C1 C2			
Actual	C1	21	6	Pred.	C1	21	6
	C2	7	41		C2	7	41

$TN/N: 41/48 = 0.854$ $41/47 = 0.872$

5. Precision
predicted total

75 Ash		Predicted	72 Ch	Actual			
		C1 C2		C1 C2			
Actual	C1	21	6	Pred.	C1	21	6
	C2	7	41		C2	7	41

$TP/(TP+FP): 21/28 = 0.750$ $21/27 = 0.778$

21 numerator
 6 numerator + these = denominator

Problem 5.5 10

QUESTION AS PRINTED 2074 Ashwin · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	25	9
	Class 2	4	31

Calculate: a) Accuracy b) Sensitivity c) Specificity d) Precision

Step 1: read the four cells

rows are Actual, so $TP = 25$, $FN = 9$, $FP = 4$, $TN = 31 \rightarrow P = 34$, $N = 35$, total 69.

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	56/69	0.812
Sensitivity	TP/P	25/34	0.735
Specificity	TN/N	31/35	0.886
Precision	$TP/(TP + FP)$	25/29	0.862

Step by step: read the axes, then each metric's cells (rows Actual)

1. Read the axes
read the axes

	Predicted	C1	C2
Actual	C1	TP 25	FN 9
	C2	FP 4	TN 31

rows Actual: $P = 34, N = 35$
KEY STEP

2. Accuracy
diagonal

	Predicted	C1	C2
Actual	C1	25	9
	C2	4	31

$(TP+TN)/all: 56/69 = 0.812$

3. Sensitivity
actual total

	Predicted	C1	C2
Actual	C1	25	9
	C2	4	31

$TP/P: 25/34 = 0.735$

4. Specificity
actual total

	Predicted	C1	C2
Actual	C1	25	9
	C2	4	31

$TN/N: 31/35 = 0.886$

5. Precision
predicted total

	Predicted	C1	C2
Actual	C1	25	9
	C2	4	31

$TP/(TP+FP): 25/29 = 0.862$

numerator
 numerator + these = denominator

Problem 5.6 3+5

QUESTION AS PRINTED 2080 Baishakh · Q3 · [3+5]

In what cases you cannot use Accuracy for performance measure, give some examples. Assume that you have the following confusion matrix. Calculate the Classification error, Sensitivity, False alarm rate Specificity.

		Actual Values	
		True	False
Predicated Values	True	1050	250
	False	150	950

Step 1: read the four cells

rows are Predicted here, so $TP = 1050, FP = 250, FN = 150, TN = 950 \rightarrow P = 1200, N = 1200$, total 2400.

Metric	Formula	Substitution	Value
Classification error	$(FP + FN)/(P + N)$	400/2400	0.167
Sensitivity	TP/P	1050/1200	0.875
False alarm rate (FPR)	FP/N	250/1200	0.208
Specificity	TN/N	950/1200	0.792

– **Check:** $FPR + specificity = 0.208 + 0.792 = 1 \checkmark$

Step by step: read the axes, then each metric's cells (rows Predicted)

1. Read the axes
read the axes

	Actual	T	F
Pred.	T	TP 1050	FP 250
	F	FN 150	TN 950

rows Pred: $P = 1200, N = 1200$
KEY STEP

2. Error rate
off-diagonal

	Actual	T	F
Pred.	T	1050	250
	F	150	950

$(FP+FN)/all: 400/2400 = 0.167$

3. Sensitivity
actual total

	Actual	T	F
Pred.	T	1050	250
	F	150	950

$TP/P: 1050/1200 = 0.875$

4. FPR
actual total

	Actual	T	F
Pred.	T	1050	250
	F	150	950

$FP/N: 250/1200 = 0.208$

5. Specificity
actual total

	Actual	T	F
Pred.	T	1050	250
	F	150	950

$TN/N: 950/1200 = 0.792$

numerator
 numerator + these = denominator

Problem 5.7 3+6

QUESTION AS PRINTED 2073 Shrawan · Q4 · [1+3+6]

Explain ROC. Using the following data, calculate TPR, FPR, precision for given confusion matrix.

	A	B
A	20	5
B	10	40

Classify, A = Yes, B = No

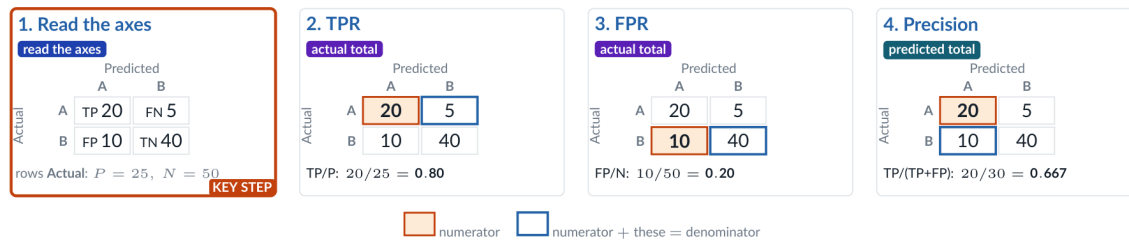
Step 1: read the four cells

A = Yes is the positive class, rows are Actual, so $TP = 20$, $FN = 5$, $FP = 10$, $TN = 40 \rightarrow P = 25$, $N = 50$.

Metric	Formula	Substitution	Value
TPR	TP/P	$20/25$	0.80
FPR	FP/N	$10/50$	0.20
Precision	$TP/(TP + FP)$	$20/30$	0.667

- The ROC point for this classifier is $(0.20, 0.80)$, comfortably above the diagonal.

Step by step: read the axes, then each metric's cells (A = Yes, rows Actual)



Problem 5.8 10

QUESTION AS PRINTED 2070 Chaitra · Q3 · [10]

An input sequence “A A B B B A A A B B” was used for classification. The Classifier ‘X’ predicted the sequences as: “A A B B B A A A B B” where as the Classifier ‘Y’ predicted the sequences as: “A A A A B B A A A B”. Develop the corresponding confusion matrix for the classifiers and find their corresponding.

- i) Accuracy ii) Precision iii) True Positive Rate iv) False Positive Rate

Step 1: tabulate Y position by position

(take A as the positive class)

Position	1	2	3	4	5	6	7	8	9	10
Actual	A	A	B	B	B	A	A	A	B	B
Y	A	A	A	A	B	B	A	A	A	B
Category	TP	TP	FP	FP	TN	FN	TP	TP	FP	TN

Step 2: the two confusion matrices, then the metrics

- X is identical to the actual sequence: $TP = 5$, $TN = 5$, $FP = 0$, $FN = 0$.
- Y from the table above: $TP = 4$, $FN = 1$, $FP = 3$, $TN = 2$; $P = 5$, $N = 5$ (total 10 ✓).

Metric	Formula	Classifier X Subst.	Value	Classifier Y Subst.	Value
Accuracy	$(TP + TN)/(P + N)$	10/10	1.00	6/10	0.60
Precision	$TP/(TP + FP)$	5/5	1.00	4/7	0.571
TPR	TP/P	5/5	1.00	4/5	0.80
FPR	FP/N	0/5	0	3/5	0.60

- X is a perfect classifier** — its ROC point is the top-left corner $(0, 1)$.
- Reading:** Y has decent recall (0.80) but poor precision (0.571) — it over-predicts A, firing 3 false alarms out of 5 actual Bs. Its ROC point $(0.60, 0.80)$ is only just above the diagonal, so it is barely better than guessing.

Step by step: tally, build both matrices, then each metric

1. Tally Y (A = positive)

tally

Actual \ Y	A	A	B	B	B	A	A	A	B	B
Y is	TP	TP	FP	FP	TN	FN	TP	TP	FP	TN

Y: $TP = 4$, $FN = 1$, $FP = 3$, $TN = 2$.
X matches Actual at every position: $TP = 5$, $TN = 5$.

KEY STEP

2. Read the axes

read the axes

Classifier X

Predicted	
A	B
Actual A	TP 5 FN 0
Actual B	FP 0 TN 5

rows Actual: $P = 5$, $N = 5$

Classifier Y

Predicted	
A	B
Actual A	TP 4 FN 1
Actual B	FP 3 TN 2

rows Actual: $P = 5$, $N = 5$

3. Accuracy

diagonal

Classifier X

Predicted	
A	B
Actual A	5 0
Actual B	0 5

(TP+TN)/all: $10/10 = 1.00$

Classifier Y

Predicted	
A	B
Actual A	4 1
Actual B	3 2

$6/10 = 0.60$

4. Precision

predicted total

Classifier X

Predicted	
A	B
Actual A	5 0
Actual B	0 5

TP/(TP+FP): $5/5 = 1.00$

Classifier Y

Predicted	
A	B
Actual A	4 1
Actual B	3 2

$4/7 = 0.571$

5. TPR

actual total

Classifier X

Predicted	
A	B
Actual A	5 0
Actual B	0 5

TP/P: $5/5 = 1.00$

Classifier Y

Predicted	
A	B
Actual A	4 1
Actual B	3 2

$4/5 = 0.80$

6. FPR

actual total

Classifier X

Predicted	
A	B
Actual A	5 0
Actual B	0 5

FP/N: $0/5 = 0$

Classifier Y

Predicted	
A	B
Actual A	4 1
Actual B	3 2

$3/5 = 0.60$

numerator
 numerator + these = denominator

Problem 5.9 4

QUESTION AS PRINTED 2079 Baishakh · Q6 · [4]

Given the following confusion matrix, determine Accuracy, Error rate, Sensitivity, Specificity, Precision, Recall of the classifier model.

$n = 165$	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

This paper labels the four cells for you, so there is no orientation trap – just read them off. $P = 105$, $N = 60$, total 165. ✓

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	$(100 + 50)/165$	0.909
Error rate	$(FP + FN)/(P + N)$	$(10 + 5)/165$	0.091
Sensitivity	TP/P	$100/105$	0.952
Specificity	TN/N	$50/60$	0.833
Precision	$TP/(TP + FP)$	$100/110$	0.909
Recall	= sensitivity	$100/105$	0.952

- **Checks:** accuracy + error rate = 1 ✓
- **Recall and sensitivity are the same thing.** The question lists both, so write both rows and say they are equal – do not assume it is a trick.
- Precision equalling accuracy here is a coincidence of the numbers, not a rule.

Step by step: read the axes, then each metric's cells (positive = YES, the second row and column)

1. Read the axes

read the axes

Predicted	
NO	YES
Actual NO	TN 50 FP 10
Actual YES	FN 5 TP 100

rows Actual: $P = 105$, $N = 60$

KEY STEP

2. Accuracy

diagonal

Predicted	
NO	YES
Actual NO	50 10
Actual YES	5 100

(TP+TN)/all: $150/165 = 0.909$

3. Error rate

off-diagonal

Predicted	
NO	YES
Actual NO	50 10
Actual YES	5 100

(FP+FN)/all: $15/165 = 0.091$

4. Sensitivity = recall

actual total

Predicted	
NO	YES
Actual NO	50 10
Actual YES	5 100

TP/P: $100/105 = 0.952$

5. Specificity

actual total

Predicted	
NO	YES
Actual NO	50 10
Actual YES	5 100

TN/N: $50/60 = 0.833$

6. Precision

predicted total

Predicted	
NO	YES
Actual NO	50 10
Actual YES	5 100

TP/(TP+FP): $100/110 = 0.909$

numerator
 numerator + these = denominator

Problem 5.10 – the mislabelled matrix 2+6

QUESTION AS PRINTED 2073 Chaitra · Q4 · [2+6]

What is ID3 algorithm? Calculate TPR, FPR and Accuracy for given confusion matrix.

	Predicted +	Predicted –
Predicted +	100	40
Predicted –	60	300

The paper labels BOTH axes “Predicted” – reproduced exactly as printed. The row stubs should read *Actual*. Say so in the answer, take rows as *Actual*, and state the assumption. That sentence is worth more than the arithmetic.

Reading rows as Actual (the intended matrix)

$TP = 100, FN = 40, FP = 60, TN = 300 \rightarrow P = 140, N = 360$, total 500.

Metric	Formula	Substitution	Value
TPR	TP/P	$100/140$	0.714
FPR	FP/N	$60/360$	0.167
Accuracy	$(TP + TN)/(P + N)$	$400/500$	0.800

If instead the rows really are Predicted

$TP = 100, FP = 40, FN = 60, TN = 300 \rightarrow P = 160, N = 340$.

Metric	Formula	Substitution	Value
TPR	TP/P	$100/160$	0.625
FPR	FP/N	$40/340$	0.118
Accuracy	$(TP + TN)/(P + N)$	$400/500$	0.800

- **Accuracy is 0.800 either way** – the diagonal does not move when you transpose. TPR and FPR do.
- **The safe exam answer:** give the first table, and add one line noting the paper’s label defect and what the numbers would become under the other reading. Full marks either way, and it shows you understand the matrix rather than a memorised formula.

Step by step: read the axes, then each metric’s cells (both readings of the mislabelled matrix)

1. Read the axes
read the axes

	Predicted			Actual				Predicted				Actual
	+		+		+		+		+		+	
Actual	+	TP 100	+	FN 40	+	TP 100	+	FP 40	+	TP 100	+	FP 40
-	-	FP 60	-	TN 300	-	FN 60	-	TN 300	-	FN 60	-	TN 300

rows Actual: $P = 140, N = 360$ rows Predicted: $P = 160, N = 340$ KEY STEP

2. TPR
actual total

	Predicted			Actual				Predicted				Actual
	+		+		+		+		+		+	
Actual	+	100	+	40	+	100	+	40	+	100	+	40
-	-	60	-	300	-	60	-	300	-	60	-	300

TP/P: $100/140 = 0.714$ 100/160 = 0.625

3. FPR
actual total

	Predicted			Actual				Predicted				Actual
	+		+		+		+		+		+	
Actual	+	100	+	40	+	100	+	40	+	100	+	40
-	-	60	-	300	-	60	-	300	-	60	-	300

FP/N: $60/360 = 0.167$ 40/340 = 0.118

4. Accuracy
diagonal

	Predicted			Actual				Predicted				Actual
	+		+		+		+		+		+	
Actual	+	100	+	40	+	100	+	40	+	100	+	40
-	-	60	-	300	-	60	-	300	-	60	-	300

(TP+TN)/all: $400/500 = 0.800$ 400/500 = 0.800

numerator
 numerator + these = denominator

M6. ANN gradients (backpropagation)

Method

- Sigmoid and its derivative: $\sigma(z) = \frac{1}{1+e^{-z}}$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.
- Loss $E = \frac{1}{2}(o - t)^2 \rightarrow \frac{\partial E}{\partial o} = (o - t)$.
- Apply the **chain rule** backwards from the loss to the weight in question.

Problem 6.1 2+3+4

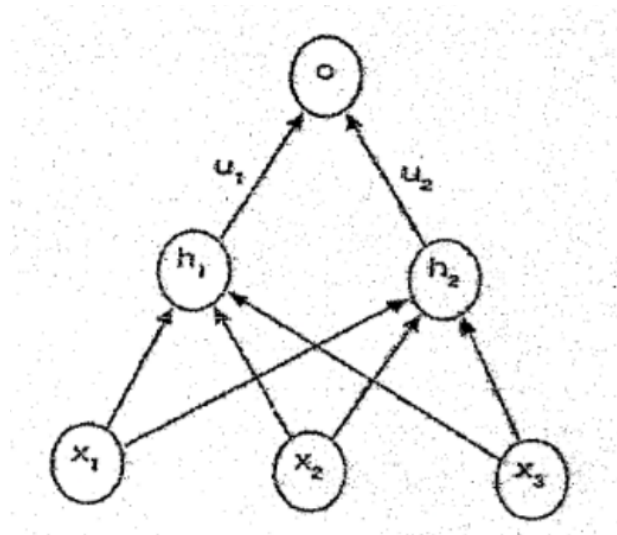
QUESTION AS PRINTED 2076 Chaitra · Q6 · [2+3+4]

Consider the multi-layer feed-forward neural network shown in the following figure. This neural network has three inputs x_1, x_2 and x_3 connected to a hidden layer consisting of two nodes h_1 and h_2 . The weight of the edge connecting x_i to h_j is w_{ji} . The two hidden nodes are connected to the output node o . The weight of the edge connecting the hidden node h_i to the output node o is u_i . The activation functions at hidden and output layers is set to sigmoid function defined as follows:

$$\sigma(\theta) = \frac{1}{1 + e^{-\theta}}$$

Using the target output t , the squared error is used as the loss function at the output node, and is defined as:

$$E(o, t) = \frac{1}{2}(o - t)^2$$



- Using the symbols given above, compute the activation at h_1 .
- Compute the gradient of the loss with respect to the output o .
- Compute the gradient of the loss with respect to the weight w_{12} .

(a) Activation at h_1

$$h_1 = \sigma(w_{11}x_1 + w_{12}x_2 + w_{13}x_3) = \frac{1}{1 + e^{-(w_{11}x_1 + w_{12}x_2 + w_{13}x_3)}}$$

(b) Gradient of the loss w.r.t. the output o

$$E(o, t) = \frac{1}{2}(o - t)^2 \Rightarrow \frac{\partial E}{\partial o} = (o - t)$$

(c) Gradient of the loss w.r.t. w_{12}

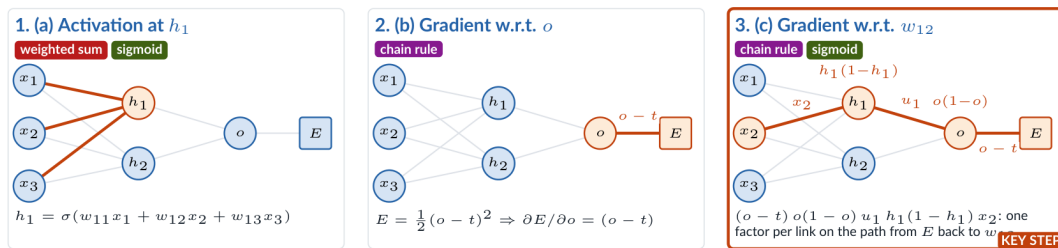
- w_{12} is the weight from x_2 into h_1 , so the chain runs $E \rightarrow o \rightarrow h_1 \rightarrow w_{12}$.
- Write $net_o = u_1h_1 + u_2h_2$ and $net_{h_1} = w_{11}x_1 + w_{12}x_2 + w_{13}x_3$.

$$\frac{\partial E}{\partial w_{12}} = \underbrace{\frac{\partial E}{\partial o}}_{(o-t)} \cdot \underbrace{\frac{\partial o}{\partial net_o}}_{o(1-o)} \cdot \underbrace{\frac{\partial net_o}{\partial h_1}}_{u_1} \cdot \underbrace{\frac{\partial h_1}{\partial net_{h_1}}}_{h_1(1-h_1)} \cdot \underbrace{\frac{\partial net_{h_1}}{\partial w_{12}}}_{x_2}$$

$$\frac{\partial E}{\partial w_{12}} = (o - t) o(1 - o) u_1 h_1(1 - h_1) x_2$$

- **Sanity checks:** the factor x_2 must appear (no input, no gradient); u_1 appears and u_2 does not, because w_{12} only influences o through h_1 .
- **Weight update:** $w_{12} \leftarrow w_{12} - l \frac{\partial E}{\partial w_{12}}$.

Step by step: forward to h_1 , then back along one path



Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity-relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	k -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.